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599 Edge AI and Robotics
Module 6 Text-to-Speech (TTS)

Overview:
This lab involved using a configured Jetson environment with docker to deploy a TTS model from the jetson-voice repository. It serves as an introduction to the TTS pipeline demonstrated in the lectures with user input text being tagged, analyzed, converted to a spectrogram, and passed through a Vocoder to a waveform to sound like speech.

Implementation:
The provided example tts.py located under jetson-voice/examples/ uses the TTS class from Jetson voice. The actual TTS class loads a TTS model, in this case the default is fastpitch_hifigan which is used with a TTSEngine to form a pipeline of two models being the the generator which generates spectrograms from tokens, and the vocoder which outputs audio from the spectrograms. The tts.py file outputs to an output device, a .wav file, or both.

Usage:
The tts.py example can be run with `./tts.py' alone, however there are optional arguments to specify an output device or where to output the .wav files. In this case the output file was chosen.

```
root@tyler-desktop:/jetson-voice# examples/tts.py --output-wav data/audio/tts_test
Namespace(debug=False, default_backend='tensorrt', global_config=None, list_devices=False, list_
models=False, log_level='info', model='fastpitch_hifigan', model_dir='/jetson-voice/data/network
s', model_manifest='/jetson-voice/data/networks/manifest.json', output_device=None, output_wav='
data/audio/tts_test', profile=False, verbose=False, warmup=5)
```

After running you simply input text and the model processes it. The generated speech goes through 5 iterations of refinement and the latency of each process time is output as well. There is nothing incredibly shocking here, the length of the generation sequence is typically positively correlated with the length of the input text, and the first iteration typically takes the most time.

```
 'defaultHighOutputLatency': 0.034829931972789115,
 'defaultLowInputLatency': 0.008707482993197279,
 'defaultLowOutputLatency': 0.008707482993197279,
 'defaultSampleRate': 44100.0,
 'hostApi': 0,
 'index': 1,
 'maxInputChannels': 16,
 'maxOutputChannels': 16,
 'name': 'tegra-snd-t210ref-mobile-rt565x: - (hw:1,0)',
 'structVersion': 2}
[2026-05-18 16:07:47] audio.py:270 - opened audio output device 1 (tegra-snd-t21
0ref-mobile-rt565x: - (hw:1,0))

Enter text, or Q to quit:
> hello

Run 0 -- Time to first audio: 4.961s. Generated 0.64s of audio. RTFx=0.13.
Run 1 -- Time to first audio: 0.354s. Generated 0.64s of audio. RTFx=1.80.
Run 2 -- Time to first audio: 0.240s. Generated 0.64s of audio. RTFx=2.66.
Run 3 -- Time to first audio: 0.235s. Generated 0.64s of audio. RTFx=2.72.
Run 4 -- Time to first audio: 0.239s. Generated 0.64s of audio. RTFx=2.67.
Run 5 -- Time to first audio: 0.251s. Generated 0.64s of audio. RTFx=2.54.
```

Each input from the user is output to a distinct .wav file, samples from running the program can be found under /media.

```
Wrote audio to data/audio/tts_test/0.wav
```